MILITARY SPECIFICATION

# SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM, LOW-POWER TYPE 2N526

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the detail requirements for a PNP, germanium, low-power transistor.

1.2 Physical dimensions. See figure 1 (TO-5).

1.3 Maximum ratings.

| $P_T$ 1/ | $V_{CBO}$ | $V_{CER}$ | $V_{EBO}$ | $I_C$ | $T_{stg}$ |
|---|---|---|---|---|---|
| mW | Vdc | Vdc | Vdc | mAdc | °C |
| 225 | -45 | -30 | -15 | -500 | -65 to +100 |

1/ Derate linearly 3 mW/°C for $T_A > 25°$ C.

1.4 Primary electrical characteristics.

| Limits | $f_{hfb}$ $V_{CB}$ = -5 Vdc $I_E$ = 1 mAdc | $h_{ob}$ $V_{CB}$ = -5 Vdc $I_E$ = 1 mAdc | $h_{rb}$ $V_{CB}$ = -5 Vdc $I_E$ = 1 mAdc | $h_{ib}$ $V_{CB}$ = -5 Vdc $I_E$ = 1 mAdc | $V_{CE}$(sat) $I_B$ = -1 mAdc $I_C$ = -20 mAdc |
|---|---|---|---|---|---|
| | MHz | μmho | $x10^{-4}$ | ohms | mVdc |
| Min | 1 | 0.1 | 1 | 25 | -55 |
| Max | 6.5 | 1 | 12 | 35 | -130 |



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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. Transistors shall be of the design, construction, and physical dimensions shown on figure 1.

3.3.1 Terminal-lead length. Terminal-lead length(s) other than that specified in figure 1 may be furnished when so stipulated under contract or order (see 6.2) where the devices covered herein are required directly for particular equipment-circuit installation or for automatic-assembly-technique programs. Where other lead lengths are required and provided, it shall not be construed as adversely affecting the qualified-product status of the device, or applicable JAN marking.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's identification.

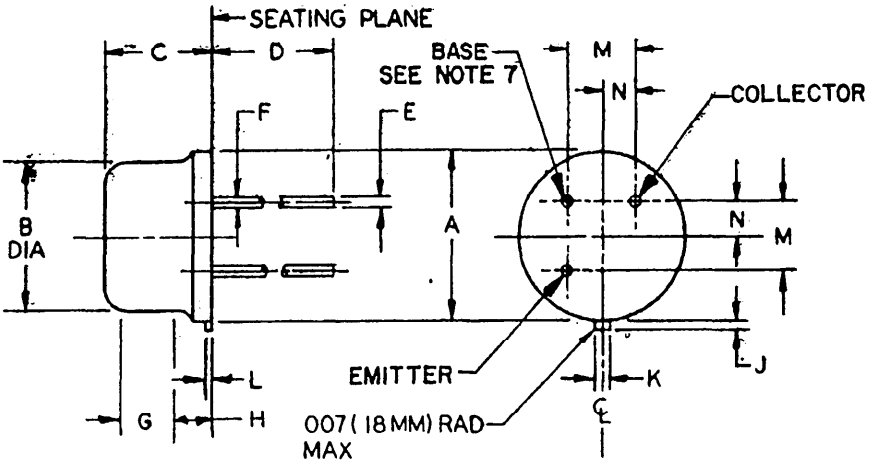
4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.



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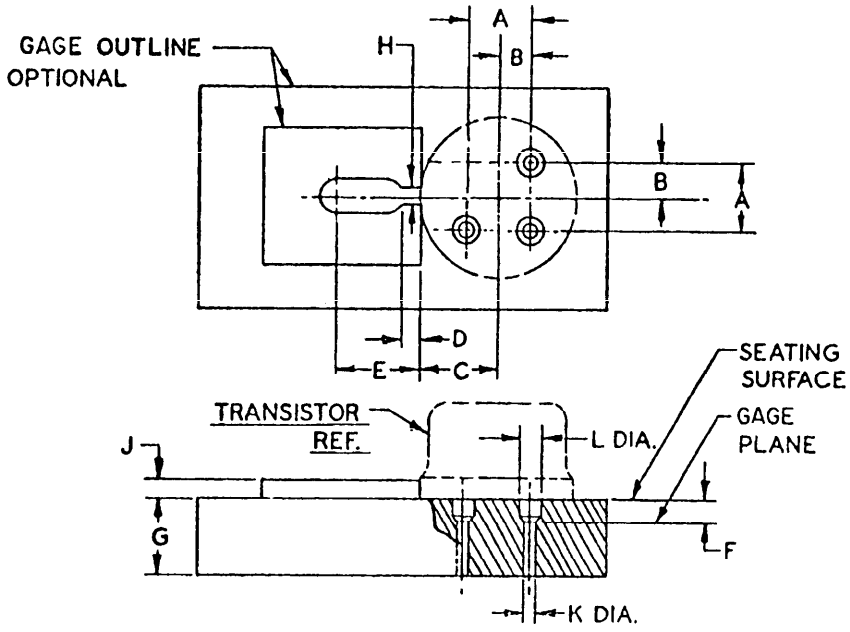

| Dimensions | | | | | |
|---|---|---|---|---|---|
| Ltr | Inches | | Millimeters | | Notes |
| | Min | Max | Min | Max | |
| A | .335 | .370 | 8.51 | 9.40 | -- |
| B | .305 | .335 | 7.75 | 8.51 | -- |
| C | .240 | .260 | 6.10 | 6.60 | -- |
| D | 1.500 | 1.750 | 38.10 | 44.45 | 9 |
| E | .016 | .021 | .41 | .53 | 2, 9 |
| F | .016 | .019 | .41 | .48 | 3, 9 |
| G | .100 | --- | 2.54 | --- | 4 |
| H | --- | --- | --- | --- | 5 |
| J | .029 | .045 | .74 | 1.14 | 8 |
| K | .028 | .034 | .71 | .86 | -- |
| L | .009 | .125 | .23 | 3.18 | -- |
| M | .1414 Nom | | 3.59 Nom | | 6 |
| N | .0707 Nom | | 1.80 Nom | | 6 |

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
3. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
4. Variations on dimension B in this zone shall not exceed .010 (.25 mm).
5. Outline in this zone is not controlled.
6. When measured in a gaging plane .054 +.001, -.000 (1.37 +.03, -.00 mm) below the seating plane of the transistor maximum diameter leads shall be within .007 (.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 2 preferred measured method.
7. The base shall be electrically connected to the case.
8. Measured from the maximum diameter of the actual device.
9. All 3 leads.

FIGURE 1. Physical dimensions of tranistor type 2N526 (TO-5).



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NOTES:
1. The following gaging procedure shall be used: The use of a pin straightener prior to insertion in the gage is permissible. The device being measured shall be inserted until its seating plane is .125 ±.010 (3.18 ±.25 mm) from the seating surface of the gage. A spacer may be used to obtain the .125 (3.18 mm) distance from the gage seat prior to force application. A force of 8 oz± 0.5 oz shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gage.
2. The location of the tab locator, within the limits of dim C, will be determined by the tab and flange dimension of the device being checked.
3. Metric equivalents ( to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

| Dimensions | | | | |
|---|---|---|---|---|
| Ltr | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | .1409 | .1419 | 3.56 | 3.60 |
| B | .0702 | .0712 | 1.78 | 1.81 |
| C | .182 | .199 | 4.62 | 5.05 |
| D | .009 | .011 | .23 | .28 |
| E | .125 Nom | | 3.18 Nom | |
| F | .054 | .055 | 1.37 | 1.40 |
| G | .372 | .378 | 9.45 | 9.60 |
| H | .0350 | .0355 | .89 | .90 |
| J | .150 Nom | | 3.81 Nom | |
| K | .0325 | .0335 | .83 | .85 |
| L | .0595 | .0605 | 1.51 | 1.54 |

**FIGURE 2.** Gage for lead and tab location for transistor type 2N526.



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4.3 Quality conformance inspection. Quality conformance inspection shall consist of group A, B, and C inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III. This inspection shall be conducted on the initial lot and thereafter every 6 months during production.

4.3.4 Group B and group C life-test samples. Samples that have been subjected to group B, 340-hours life-test, may be continued on test to 1,000 hours in order to satisfy group C life-test requirements. These samples shall be predesignated, and shall remain subjected to the group C 1,000-hour acceptance evaluation after they have passed the group B, 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1,000 hours shall be computed for 1,000-hour acceptance criteria.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II,and III.

TABLE I. Group A inspection

| Examination or test | MIL-STD-750 | | LTPD | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | Method | Details | | | Min | Max | |
| Subgroup 1 | | | 10 | | | | |
| Visual and mechanical examination | 2071 | | | --- | --- | --- | --- |
| Subgroup 2 | | | 5 | | | | |
| Breakdown voltage, collector to emitter | 3011 | Bias cond. B; $I_C$ = -0.6 mAdc; $R_2$ = 10 K | | $BV_{CER}$ | -30 | --- | Vdc |
| Breakdown voltage, collector to base | 3001 | Bias cond. D; $I_C$ = -200 μAdc | | $BV_{CBO}$ | -45 | --- | Vdc |
| Breakdown voltage, emitter to base | 3026 | Bias cond. D; $I_E$ = -10 μAdc | | $BV_{EBO}$ | -15 | --- | Vdc |
| Collector to base cutoff current | 3036 | Bias cond. D; $V_{CB}$ = -30 Vdc | | $I_{CBO}$ | --- | -10 | μAdc |
| Floating potential | 3020 | $V_{CB}$ = -31 Vdc; Voltmeter input resistance $\geq$ 10 megohms | | $V_{EBF}$ | --- | -1 | Vdc |



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TABLE I. Group A inspection - Continued

| Examination or test | MIL-STD-750 | | LTPD | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | Method | Details | | | Min | Max | |
| Subgroup 3 | | | 5 | | | | |
| Small-signal open-circuit output admittance | 3216 | $V_{CB}$ = -5 Vdc; $I_E$ = 1 mAdc | | $h_{ob}$ | 0.1 | 1 | μmho |
| Small-signal open-circuit reverse-voltage transfer ratio | 3211 | $V_{CB}$ = -5 Vdc; $I_E$ = 1 mAdc | | $h_{rb}$ | 1 | 12 | x10-4 |
| Small-signal short-circuit input impedance | 3201 | $V_{CB}$ = -5 Vdc; $I_E$ = 1 mAdc | | $h_{ib}$ | 25 | 35 | ohms |
| Small-signal short-circuit forward-current transfer ratio | 3206 | $V_{CB}$ = -5 Vdc; $I_E$ = 1 mAdc | | $h_{fe}$ | 44 | 88 | --- |
| Subgroup 4 | | | 10 | | | | |
| Open-circuit output capacitance | 3236 | $V_{CB}$ = -5 Vdc; $I_E$ = 0; 100 kHz $\leq$ f $\leq$ 1 MHz | | $C_{obo}$ | 10 | 40 | pf |
| Noise figure | 3246 | $V_{CB}$ = -5 Vdc; $I_E$ = 1 mAdc; Rg = 1, 000 ohms | | NF | --- | 10 | db |
| Small-signal short-circuit forward-current transfer-ratio cutoff frequency | 3301 | $V_{CB}$ = -5 Vdc; $I_E$ = 1 mAdc | | $f_{hfb}$ | 1 | 6.5 | MHz |
| Subgroup 5 | | | 5 | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -1 Vdc; $I_C$ = -20 mAdc | | $h_{FE}$ | 53 | 90 | --- |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -1 Vdc; $I_C$ = -100 mAdc | | $h_{FE}$ | 47 | --- | --- |
| Base emitter voltage (nonsaturated) | 3066 | Test cond. B; $V_{CE}$ = -1 Vdc; $I_C$ = -20 mAdc | | $V_{BE}$ | -190 | -320 | mVdc |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = -20 mAdc; $I_B$ = -1 mAdc | | $V_{CE}(sat)$ | -55 | -130 | mVdc |
| Subgroup 6 | | | 15 | | | | |
| High-temperature operation: | | $T_A$ = 70°C | | | | | |
| Collector to base cutoff current | 3036 | Bias cond. D; $V_{CB}$ = -30 Vdc | | $I_{CBO}$ | --- | -200 | μAdc |
| Low-temperature operation: | | $T_A$ = -55°C | | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -1 Vdc; $I_C$ = -20 mAdc | | $h_{FE}$ | 27 | --- | --- |



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TABLE II. Group B inspection

| Examination or test | MIL-STD-750 | | LTPD | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | Method | Details | | | Min | Max | |
| Subgroup 1 | | | 20 | | | | |
| Physical dimensions | 2066 | (See figure 1) | | --- | --- | --- | --- |
| Subgroup 2 | | | 15 | | | | |
| Solderability | 2026 | | | --- | --- | --- | --- |
| Thermal shock (temperature cycling) | 1051 | Test cond. B, except step 3, $T_A$ = +100°C | | --- | --- | --- | --- |
| Thermal shock (glass strain) | 1056 | Test cond. A | | --- | --- | --- | --- |
| Hermetic seal | 1071 | Test cond. G or H for fine leak tests; test cond. A, C, D or F for gross leaks | | --- | --- | $1x10^{-7}$ | atm cc/sec |
| Moisture resistance | 1021 | | | --- | --- | --- | --- |
| End points: | | | | | | | |
| Collector to base cutoff current | 3036 | Bias cond. D; $V_{CB}$ = -30 Vdc | | $I_{CBO}$ | --- | -15 | µAdc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -1 Vdc; $I_C$ = -20 mAdc | | $h_{FE}$ | 50 | 100 | --- |
| Subgroup 3 | | | 15 | | | | |
| Shock | 2016 | Nonoperating; 1,500 G, 0.5 msec, 5 blows in each orientation: $X_1$, $Y_1$, $Y_2$, and $Z_1$ | | --- | --- | --- | --- |
| Vibration, variable frequency | 2056 | | | --- | --- | --- | --- |
| Constant acceleration | 2006 | 10,000 G in each orientation: $X_1$, $Y_1$, $Y_2$, and $Z_1$ | | --- | --- | --- | --- |
| End points: (Same as subgroup 2) | | | | | | | |
| Subgroup 4 | | | 20 | | | | |
| Terminal strength (lead fatigue) | 2036 | Test cond. E | | --- | --- | --- | --- |
| End points: | | | | | | | |
| Hermetic seal | 1071 | Test cond. G or H for fine leaks; test cond. A, C, D, or F for gross leaks | | --- | --- | $1x10^{-7}$ | atm cc/sec |



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TABLE II. Group B inspection - Continued

| Examination or test | MIL-STD-750 | | LTPD | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | Method | Details | | | Min | Max | |
| Subgroup 5 | | | 20 | | | | |
| Salt atmosphere (corrosion) | 1041 | | | --- | --- | --- | --- |
| End points: (Same as subgroup 2) | | | | | | | |
| Subgroup 6 | | | 7 | | | | |
| High-temperature life (nonoperating) | 1032 | $T_{stg}$ = +100°C; time = 340 hours (see 4.3.4) | | --- | --- | --- | --- |
| End points: | | | | | | | |
| Collector to base cutoff current | 3036 | Bias cond. D; $V_{CB}$ = -30 Vdc | | $I_{CBO}$ | --- | -20 | µAdc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -1 Vdc; $I_C$ = -20 mAdc | | $h_{FE}$ | 45 | 110 | --- |
| Subgroup 7 | | | 7 | | | | |
| Intermittent operation life | 1037 | $V_{CB}$ = -20 Vdc; $P_C$ = 225 mW; Cycle: ON, 50 ±2 minutes, OFF, 10 ±2 minutes; time = 340 hours (see 4.3.4) | | --- | --- | --- | --- |
| End points: (Same as subgroup 6) | | | | | | | |

TABLE III. Group C inspection

| Examination or test | MIL-STD-750 | | LTPD | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | Method | Details | | | Min | Max | |
| Subgroup 1 | | | λ = 7 | | | | |
| High-temperature life (nonoperating) | 1031 | $T_{stg}$ = +100°C (see 4.3.4) | | --- | --- | --- | --- |
| End points: (Same as subgroup 6 of group B) | | | | | | | |
| Subgroup 2 | | | λ = 7 | | | | |
| Intermittent operation life | 1036 | $V_{CB}$ = -20 Vdc; $P_C$ = 225 mW; Cycle: ON, 50 ±2 minutes, OFF, 10 ±2 minutes (see 4.3.4) | | --- | --- | --- | --- |
| End points: (Same as subgroup 6 of group B) | | | | | | | |



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5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500, section 5.

6. NOTES:

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data. Procurement documents should specify the following: Terminal-lead length if other than specified in figure 1 (see 3.3.1).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU, MI
Navy - SH
Air Force - 11, 17
DSA - ES

User activities:
Army - SM
Navy - CG, MC, OS, AS
Air Force - 13, 15, 19, 70, 80

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5961-0159)



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# SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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